

Zeqian Long

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RESEARCH STATEMENT

My research focuses on **generative AI**, with an emphasis on **controllability** and **reliability**. I work on controllable visual synthesis, robustness evaluation, and structured knowledge integration—capable of modeling the world’s structure and dynamics in precise, trustworthy, and grounded ways.

EDUCATION

May 2026	University of Illinois Urbana–Champaign	IL, United States
Aug 2022	B.S. in Mathematics and Computer Science (Dean’s List)	GPA: 4.0 / 4.0
	Coursework (* = graduate-level): Deep Generative Models*, Text Analysis with LLMs*, Graph Learning & Knowledge Mining*, Statistical Machine Learning*, Advanced Algorithms*, Theoretical Mathematics (Stochastic Process, Real Analysis, Abstract Algebra)	

PUBLICATIONS

S=IN SUBMISSION (* = EQUAL CONTRIBUTION)

- [S.1] **Follow-Your-Shape: Shape-Aware Image Editing via Trajectory-Guided Region Control**
Zeqian Long*, Mingzhe Zheng*, Kunyu Feng*, Xinhua Zhang, Hongyu Liu, Harry Yang, Linfeng Zhang, Qifeng Chen, Yue Ma
Under Review at ICLR 2026 (scores: 6, 6, 4, 6) [[arXiv](#) | [GitHub](#) | [OpenReview](#)]

RESEARCH EXPERIENCES

Shape-Aware Image Editing via Trajectory-Guided Region Control MAR’25 – SEP’25

EPIC Lab, Shanghai Jiao Tong University, supervised by *Yue Ma, Prof. Linfeng Zhang*

- › Developed a training- and mask-free image editing framework built on FLUX.1-dev that dynamically localizes editable areas and enables precise and large-scale shape transformations
- › Demonstrated state-of-the-art performance on PIE-Bench and our constructed 290-case ReShapeBench, outperforming FLUX.1 Kontext by over 10% across key quality and fidelity metrics.
- › First author; in submission to **ICLR 2026**.

Attention-Driven Adversarial Attack for Image-To-Video Models JUL’25 – PRESENT

UIUC, collaborated with *Haotian Xue, Ozgur Kara*

- › Proposed a memory-efficient adversarial attack framework for the Image-to-Video model **Wan2.1-14B**, inducing significant drops in semantic consistency and degradation in temporal coherence and visual fidelity.
- › Designed a cross-attention-driven semantic loss that contrasts source-target CLIP and latent attention to extract semantic deviation signals; integrated multi-resolution VAE encoder feature loss that amplifies reconstruction errors across scales; applied PGD to optimize adversarial perturbations on input images.
- › **Github**: <https://github.com/Zeqian-Long/Wan-I2V-Attack>; **Target**: On track for **ICML 2026** submission.

Ontology-Grounded Scientific Graph Construction for LLM Retrieval and Reasoning AUG’25 – PRESENT

ULab, UIUC, supervised by *Prof. Jiaxuan You, Prof. Jiawei Han*

- › Constructed a hierarchical scientific ontology that decomposes papers across task, method, dataset, architecture, evaluation, and contribution dimensions to capture detailed structural semantics.
- › Developed a paper-to-ontology mapping method that assigns each paper to specific concepts with explicit roles (uses / introduces), producing interpretable concept-level representations.
- › Identified concept-induced paper relations that reveal deeper semantic connections beyond citations, directly improving LLM fine-grained retrieval, multi-hop scientific QA, and cross-paper comparison analysis.
- › **Target**: On track for **ARR/ACL 2026** submission.

Neuro-Symbolic Knowledge Graph for Aligning LLM Worldviews AUG’25 – PRESENT

UIUC, supervised by *Prof. Jiawei Han*

- > Constructed a neuro-symbolic framework that constructs a Global Values Knowledge Graph by integrating quantitative survey data (WVS, EVS), legal texts, and cultural corpora from 50+ countries to support verifiable cross-cultural and cross-lingual alignment of LLM worldviews.
- > Proposed a Graph-of-Values reasoning protocol together with diagnostic metrics that assess (i) the accuracy of an LLM's reasoning paths over value questions, (ii) the coherence and grounding of concept-level answers, and (iii) the model's sensitivity to temporal shifts in global value trends.

PROJECTS

SmartRide – Smart Cycling App for Smarter Riders JAN'25 – MAY'25

Course Project, full-stack development

- > Designed a smart cycling platform integrating route planning, real-time tracking, and personalized user profiling.
- > Implemented a route planning module with dynamic map generation, distance optimization, and turn-by-turn navigation; built a route persistence system enabling save, edit, and reload of user-defined paths.
- > **Github:** <https://github.com/CS222-UIUC/team-07-project-smartride>

Predicting Crop Yield based on Sentinel-2 Satellite Data JAN'25 – MAY'25

Independent Project

- > Preprocessed Sentinel-2 satellite imagery by aligning it with European crop type maps, sampling consistent spatial blocks, and normalizing spectral features across growth periods.
- > Applied modern dimensionality reduction to convert preprocessed images into histograms; trained a CNN and employed Deep Gaussian Process to extract key spectral-temporal patterns for preliminary crop yield estimation across 500 regional samples.

WORKING EXPERIENCES

Dec 2024	UIUC Teaching Assistanship CS 446: Machine Learning	IL, United States
Aug 2024	<i>Course Assistant — Professor: Liangyan Gui</i> > Assist students with machine learning concepts; proofread and grade the exams; manage office hours; support course operations to enhance learning outcomes.	
Aug 2024	Beijing Academy of Artificial Intelligence Vision and Multimodal Research Center	Beijing, CN
Jun 2024	<i>Research Intern — Advisor: Xinlong Wang</i> > Reproduced diffusion and autoregressive image generation papers, examining architectural decisions and organizing experimental workflows. > Contributed to multimodal visual generation research within the Emu framework through baseline models evaluations, cross-modal generation quality assessments, and experiment documentations.	

SKILLS

Programming	Python (PyTorch, TensorFlow), C/C++, Java, Javascript, Bash
AI/ML Frameworks	PyTorch Lightning, HuggingFace Transformers, Diffusers, OpenCV, scikit-learn
Systems/Tools	Linux, Git, Docker, CUDA Toolkit